

CORRESPONDENCE



Driving pressure: pitfalls of methods that do not use esophageal pressure measurement

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I have read the article by Oriol Roca et al. [1] and appreciate the authors' efforts to identify a method for bedside assessment of driving pressure. This method allowed many clinicians to easily assess lung strain without esophageal pressure measurement. While I totally agree with the authors' opinion, I'm concerned that the limitations of these methods were not mentioned.

First, the authors should mention that transpulmonary pressure (ΔP_L) is considered a more important indicator than driving pressure (ΔP). The authors present a study in which ΔP did not significantly differ from ΔP_L in 60-day mortality in limitation [2], and they show that thoracic elastance may be related to disease severity and that changes in ΔP may assess disease severity. Considering the fact that their study was an observational study and the concept of strain [3], I believe that ΔP_L is more appropriate indicator to prevent VILI than ΔP , and that an accurate method for evaluating ΔP_L is necessary, as described below.

Second, the coefficient 0.66 of occlusion pressure (P_{occ}) to determine ΔP_L is not constant in all patients. Transpulmonary pressure is determined by the ratio of chest wall and respiratory system elastance. Because this ratio varies from patient to patient, Gattinoni et al. noted in their article that it was impossible to know whether airway pressure is harmful to a patient without measuring esophageal pressure [4]. It is difficult to apply the coefficient 0.66 to patients who are evaluated for ΔP_L to

prevent ventilator-induced lung injury (VILI), because the values of this ratio are diverse.

Third, the peak pressure obtained by inspiratory effort and the ventilator may not coincide in time and phase. Therefore, the method of summing occlusion pressure and airway pressure separately may overestimate driving pressure. Especially in patients with high airway resistance, there is a time lag between the start of inspiration and the peak alveolar pressure [5], and may mislead the clinicians to recognize the peak time of alveolar pressure (Fig. 1).

Gattinoni et al. noted that esophageal pressure should be measured even if it increases workload for doctors and nurses [4]. Although a simple bedside measurement method is important, we believe that accurate assessment

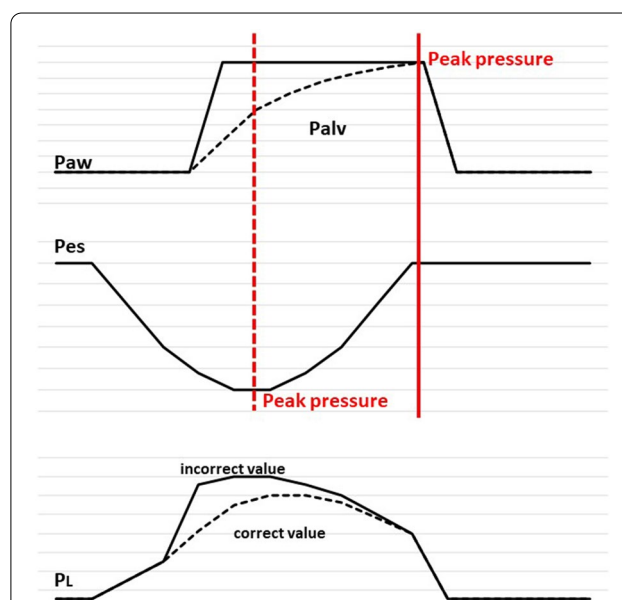


Fig. 1 If we determine the strain of the lungs, we need to consider the time phase of the alveolar pressure (Palv) and esophageal pressure (Pes). Adding the peaks of each may overestimate the transpulmonary pressure (PL) if the time phases do not coincide

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of transpulmonary pressure by esophageal pressure measurement is more relevant to patient outcome.

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Declarations

Conflict of interest

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